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Roll No

EC-702(B)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Information Theory and Coding

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss uncertainty, entropy with its properties as per theory of information.
b) Derive the Channel Capacity of Binary Erasure Channel.
2. A very noisy symmetric binary channel is able to transmit 1000 bits per second, but has a 25% probability of introducing error into each bit transmitted. What is the channel capacity in bits per second?
3. a) Show that the mutual information $I(X; Y) \leq 0$. Under what condition does the equality hold?
b) Show that mutual information is symmetric.
4. a) Prove that the average code-word length $E(w)$ of a binary instantaneous code for discrete memory less source with entropy $H(X)$ satisfies the inequality $H(X) \leq E(W) < H(X) + 1$.
b) Derive the channel capacity of band limited Gaussian channels.

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5. a) State and prove Kraft inequality.
b) Give example for a Hamming code. Write down generator matrix and parity check matrix of that code. Comment on the error correction capability of the Hamming Code.
6. a) Explain generation of systematic cyclic code.
b) The Viterbi algorithm used for decoding of convolution codes is an example of Maximum likelihood decoding. Explain.
7. a) Obtain the standard array of the linear block code with code words $C = \{0000, 1001, 0110, 1111\}$.
b) What are different types of errors? Explain them with suitable examples.
8. a) Write brief notes on
 - i) Trellis and state diagrams
 - ii) Channel Models and its Matrix
b) Comment on the following: Lempel-Ziv Coding , Error correction capabilities of linear block codes.
